Windows CIFS/SMB Share Permissions – Collection, Comparison, and Validation Guide

Practical documentation for collecting permissions from old and new CIFS shares, comparing NTFS ACLs and share ACLs, identifying missing data, and automating the process with PowerShell.

1. Objective

This guide shows how to validate a Windows CIFS/SMB migration or copy by comparing the old share and the new share. It covers the three layers you should check:
• **File/folder inventory** – to find missing or extra files/directories
• **NTFS file/folder ACLs** – to compare permissions inside the share
• **SMB share permissions** – to compare the permissions configured on the share itself

2. What exactly needs to be compared

For a reliable validation, compare these separately:

A. Inventory / data presence
Confirms whether all files and folders copied from the old share to the new share.

B. NTFS permissions on files and folders
These are the ACLs on the objects inside the share, such as Read, Modify, Full Control, inheritance, owner, and allow/deny entries.

C. SMB share permissions
These are the permissions configured on the share itself, such as Everyone=Read or Domain Admins=Full.

3. Important distinction: Share ACL vs NTFS ACL

A CIFS/SMB share has two permission layers:
• **Share permissions**: who can access the share over SMB
• **NTFS ACLs**: what a user can do to the files and folders inside the share

Effective access is the combination of both. That is why a proper migration validation should compare both.

4. Recommended validation workflow

Use the following order:
1) Compare the file/folder inventory
2) Compare NTFS ACLs on the files and directories
3) Compare SMB share permissions
4) Review mismatches and confirm whether they are expected or not

5. Prerequisites

Run the scripts from a Windows host or jump server that can access both the old and new shares.

Requirements:
• PowerShell 5.1 or later
• Read access to both old and new shares
• Permission to read ACLs on the shares
• For share-level comparison with Get-SmbShareAccess, administrative access on the file server that hosts the share is usually required
• Enough free disk space to store CSV output if the shares are large

6. Example environment used in this guide

```
Old share path : \\oldserver\finance
New share path : \\newserver\finance
Share name     : finance
```

7. Step 1 – Collect inventory from the old and new shares

This tells you whether files and folders are missing, extra, or of a different type (file vs directory).

7.1 Old share inventory

```
$OldShare = "\\oldserver\finance"

Get-ChildItem $OldShare -Recurse -Force | Select-Object `
    @{Name='Path';Expression={$_.FullName.Replace($OldShare,'')}},
    @{Name='ItemType';Expression={if ($_.PSIsContainer) {'Directory'} else {'File'}}}
| Export-Csv "C:\Temp\old_inventory.csv" -NoTypeInformation
```

7.2 New share inventory

```
$NewShare = "\\newserver\finance"

Get-ChildItem $NewShare -Recurse -Force | Select-Object `
    @{Name='Path';Expression={$_.FullName.Replace($NewShare,'')}},
    @{Name='ItemType';Expression={if ($_.PSIsContainer) {'Directory'} else {'File'}}}
| Export-Csv "C:\Temp\new_inventory.csv" -NoTypeInformation
```

7.3 Compare inventory

```
$oldInv = Import-Csv "C:\Temp\old_inventory.csv"
$newInv = Import-Csv "C:\Temp\new_inventory.csv"

Compare-Object `
    -ReferenceObject $oldInv `
    -DifferenceObject $newInv `
    -Property Path,ItemType `
    -PassThru |
Export-Csv "C:\Temp\inventory_diff.csv" -NoTypeInformation
```

Interpretation:
- **<=** means present in old share but missing or different in new share
- **=>** means present in new share but not in old share

8. Step 2 – Collect NTFS permissions (ACLs) from the old and new shares

This is the main permission comparison. For every file and folder, the script collects:
- relative path
- item type (file/directory)
- owner
- identity reference (user/group)
- FileSystemRights
- Allow/Deny
- inheritance flags
- propagation flags
- whether the ACE is inherited

8.1 Old share NTFS ACL export

```
$OldShare = "\\oldserver\finance"

Get-ChildItem $OldShare -Recurse -Force | ForEach-Object {
    $acl = Get-Acl $_.FullName
    foreach ($ace in $acl.Access) {
        [PSCustomObject]@{
            Path               = $_.FullName.Replace($OldShare, "")
```

```
            ItemType           = if ($_.PSIsContainer) { "Directory" } else { "File" }
            Owner              = $acl.Owner
            IdentityReference  = $ace.IdentityReference
            FileSystemRights   = $ace.FileSystemRights
            AccessControlType  = $ace.AccessControlType
            InheritanceFlags   = $ace.InheritanceFlags
            PropagationFlags   = $ace.PropagationFlags
            IsInherited        = $ace.IsInherited
        }
    }
} | Export-Csv "C:\Temp\old_ntfs_acl.csv" -NoTypeInformation
```

8.2 New share NTFS ACL export

```
$NewShare = "\\newserver\finance"

Get-ChildItem $NewShare -Recurse -Force | ForEach-Object {
    $acl = Get-Acl $_.FullName
    foreach ($ace in $acl.Access) {
        [PSCustomObject]@{
            Path               = $_.FullName.Replace($NewShare, "")
            ItemType           = if ($_.PSIsContainer) { "Directory" } else { "File" }
            Owner              = $acl.Owner
            IdentityReference  = $ace.IdentityReference
            FileSystemRights   = $ace.FileSystemRights
            AccessControlType  = $ace.AccessControlType
            InheritanceFlags   = $ace.InheritanceFlags
            PropagationFlags   = $ace.PropagationFlags
            IsInherited        = $ace.IsInherited
        }
    }
} | Export-Csv "C:\Temp\new_ntfs_acl.csv" -NoTypeInformation
```

8.3 Compare NTFS ACL exports

```
$old = Import-Csv "C:\Temp\old_ntfs_acl.csv"
$new = Import-Csv "C:\Temp\new_ntfs_acl.csv"

Compare-Object `
    -ReferenceObject $old `
    -DifferenceObject $new `
    -Property Path,ItemType,Owner,IdentityReference,FileSystemRights,AccessControlType,InheritanceFlags,Propagatio
    -PassThru |
Export-Csv "C:\Temp\ntfs_acl_diff.csv" -NoTypeInformation
```

What this catches:

• Missing ACEs on the new share
• Extra ACEs on the new share
• Different rights such as Read vs Modify
• Owner changes
• Inheritance differences
• Allow vs Deny differences

9. Step 3 – Collect and compare SMB share permissions

Share-level permissions are separate from NTFS ACLs. Compare them too.

9.1 Old share ACL export

```
Get-SmbShareAccess -Name "finance" | Export-Csv "C:\Temp\old_share_acl.csv" -NoTypeInformation
```

9.2 New share ACL export

```
Get-SmbShareAccess -Name "finance" | Export-Csv "C:\Temp\new_share_acl.csv" -NoTypeInformation
```

9.3 Compare share ACLs

```
$oldShareAcl = Import-Csv "C:\Temp\old_share_acl.csv"
$newShareAcl = Import-Csv "C:\Temp\new_share_acl.csv"
```

```
Compare-Object `
    -ReferenceObject $oldShareAcl `
    -DifferenceObject $newShareAcl `
    -Property Name,AccountName,AccessControlType,AccessRight `
    -PassThru |
Export-Csv "C:\Temp\share_acl_diff.csv" -NoTypeInformation
```

10. Reading the results

After running the three comparisons, you will typically have these output files:

inventory_diff.csv
Shows files/directories missing in the new share or unexpectedly present in the new share.

ntfs_acl_diff.csv
Shows differences in NTFS permissions, ownership, inheritance, and ACE entries.

share_acl_diff.csv
Shows differences in the SMB share permissions.

11. Practical interpretation examples

Example 1 – Missing file in the new share

If inventory_diff.csv shows a path only on the old share, the data copy is incomplete for that item.

Example 2 – Permission mismatch

If ntfs_acl_diff.csv shows the same path but different FileSystemRights, the object exists in both places but the permissions are not the same.

Example 3 – Inheritance mismatch

If IsInherited or InheritanceFlags differ, the ACL structure changed during the copy or was inherited differently on the new share.

Example 4 – Share permission mismatch

If share_acl_diff.csv shows Everyone=Read on old but not on new, users may not even be able to reach the share despite NTFS permissions being correct.

12. Common causes of differences

Common reasons for mismatches:
• The copy tool did not preserve NTFS ACLs
• The copy preserved data but not owner/SACL/DACL information
• Inheritance was re-applied on the target
• Share permissions were recreated manually and do not match
• Some files were skipped due to long path, open file, or access issues
• The old and new shares were exported from different parent folders with inherited ACL differences

13. Recommended copy tools and flags

If you are copying CIFS data and want to preserve NTFS permissions, choose a tool and flags that preserve security information. Two common approaches are:

13.1 Robocopy example

```
robocopy "\\oldserver\finance" "\\newserver\finance" /E /COPYALL /DCOPY:DAT /R:2 /W:2 /MT:16 /LOG:C:\Temp\robocopy
```

Useful robocopy flags:

- **/E** – copy subdirectories including empty ones
- **/COPYALL** – copy data, attributes, timestamps, NTFS ACLs, owner, and auditing info
- **/DCOPY:DAT** – preserve directory data/attributes/timestamps
- **/R** and **/W** – retry tuning
- **/LOG** – write a log for troubleshooting

14. Single end-to-end PowerShell script

The following script collects inventory, NTFS ACLs, and share ACLs for both the old and new share, then compares them and writes all outputs to a timestamped folder.

```
param(
    [string]$OldShare = "\\oldserver\finance",
    [string]$NewShare = "\\newserver\finance",
    [string]$OldShareName = "finance",
    [string]$NewShareName = "finance",
    [string]$OutputRoot = "C:\Temp"
)

$TimeStamp = Get-Date -Format "yyyyMMdd_HHmmss"
$OutDir = Join-Path $OutputRoot "CIFS_Compare_$TimeStamp"
New-Item -Path $OutDir -ItemType Directory -Force | Out-Null

Write-Host "Output directory: $OutDir"

# --------------------------
# 1) Inventory collection
# --------------------------
Write-Host "Collecting inventory from old share..."
Get-ChildItem $OldShare -Recurse -Force | Select-Object `
    @{Name='Path';Expression={$_.FullName.Replace($OldShare,'')}},
    @{Name='ItemType';Expression={if ($_.PSIsContainer) {'Directory'} else {'File'}}} |
Export-Csv (Join-Path $OutDir "old_inventory.csv") -NoTypeInformation

Write-Host "Collecting inventory from new share..."
Get-ChildItem $NewShare -Recurse -Force | Select-Object `
    @{Name='Path';Expression={$_.FullName.Replace($NewShare,'')}},
    @{Name='ItemType';Expression={if ($_.PSIsContainer) {'Directory'} else {'File'}}} |
Export-Csv (Join-Path $OutDir "new_inventory.csv") -NoTypeInformation

$oldInv = Import-Csv (Join-Path $OutDir "old_inventory.csv")
$newInv = Import-Csv (Join-Path $OutDir "new_inventory.csv")

Compare-Object -ReferenceObject $oldInv -DifferenceObject $newInv `
    -Property Path,ItemType -PassThru |
Export-Csv (Join-Path $OutDir "inventory_diff.csv") -NoTypeInformation

# --------------------------
# 2) NTFS ACL collection
# --------------------------
function Export-NtfsAcl {
    param(
        [string]$SharePath,
        [string]$OutFile
    )

    Get-ChildItem $SharePath -Recurse -Force | ForEach-Object {
        try {
            $acl = Get-Acl $_.FullName
            foreach ($ace in $acl.Access) {
                [PSCustomObject]@{
                    Path              = $_.FullName.Replace($SharePath, "")
                    ItemType          = if ($_.PSIsContainer) { "Directory" } else { "File" }
                    Owner             = $acl.Owner
                    IdentityReference  = $ace.IdentityReference
                    FileSystemRights   = $ace.FileSystemRights
                    AccessControlType  = $ace.AccessControlType
                    InheritanceFlags   = $ace.InheritanceFlags
                    PropagationFlags   = $ace.PropagationFlags
                    IsInherited        = $ace.IsInherited
                }
```

```
            }
        } catch {
            [PSCustomObject]@{
                Path               = $_.FullName.Replace($SharePath, "")
                ItemType           = if ($_.PSIsContainer) { "Directory" } else { "File" }
                Owner              = "ERROR"
                IdentityReference  = "ERROR"
                FileSystemRights   = $_.Exception.Message
                AccessControlType  = ""
                InheritanceFlags   = ""
                PropagationFlags   = ""
                IsInherited        = ""
            }
        }
    } | Export-Csv $OutFile -NoTypeInformation
}

Write-Host "Collecting NTFS ACLs from old share..."
Export-NtfsAcl -SharePath $OldShare -OutFile (Join-Path $OutDir "old_ntfs_acl.csv")

Write-Host "Collecting NTFS ACLs from new share..."
Export-NtfsAcl -SharePath $NewShare -OutFile (Join-Path $OutDir "new_ntfs_acl.csv")

$oldAcl = Import-Csv (Join-Path $OutDir "old_ntfs_acl.csv")
$newAcl = Import-Csv (Join-Path $OutDir "new_ntfs_acl.csv")

Compare-Object -ReferenceObject $oldAcl -DifferenceObject $newAcl `
    -Property Path,ItemType,Owner,IdentityReference,FileSystemRights,AccessControlType,InheritanceFlags,Propagatio
    -PassThru |
Export-Csv (Join-Path $OutDir "ntfs_acl_diff.csv") -NoTypeInformation

# -------------------------
# 3) Share ACL collection
# -------------------------
Write-Host "Collecting old share ACL..."
Get-SmbShareAccess -Name $OldShareName |
Export-Csv (Join-Path $OutDir "old_share_acl.csv") -NoTypeInformation

Write-Host "Collecting new share ACL..."
Get-SmbShareAccess -Name $NewShareName |
Export-Csv (Join-Path $OutDir "new_share_acl.csv") -NoTypeInformation

$oldShareAcl = Import-Csv (Join-Path $OutDir "old_share_acl.csv")
$newShareAcl = Import-Csv (Join-Path $OutDir "new_share_acl.csv")

Compare-Object -ReferenceObject $oldShareAcl -DifferenceObject $newShareAcl `
    -Property Name,AccountName,AccessControlType,AccessRight -PassThru |
Export-Csv (Join-Path $OutDir "share_acl_diff.csv") -NoTypeInformation

# -------------------------
# 4) Summary
# -------------------------
$summary = @()
$summary += "Output directory: $OutDir"
$summary += "Files generated:"
$summary += " - old_inventory.csv"
$summary += " - new_inventory.csv"
$summary += " - inventory_diff.csv"
$summary += " - old_ntfs_acl.csv"
$summary += " - new_ntfs_acl.csv"
$summary += " - ntfs_acl_diff.csv"
$summary += " - old_share_acl.csv"
$summary += " - new_share_acl.csv"
$summary += " - share_acl_diff.csv"

$summary | Set-Content (Join-Path $OutDir "summary.txt")

Write-Host "Done. Review the output under $OutDir"
```

15. Notes for large shares

For very large shares, keep these points in mind:

• ACL exports can be large; write to CSV files rather than console

• Run during a quiet period if possible
• If the share contains millions of files, the process can take a long time
• Consider splitting the run per top-level folder if the share is extremely large
• Keep the output directory on a drive with enough free space

16. Troubleshooting tips

If you get access denied:
• Make sure the account running the script has read access to the share and permission to read ACLs
• Run PowerShell as an account with sufficient rights
• For Get-SmbShareAccess, run on the file server or with administrative rights if required

If the script is slow:
• Test on one subfolder first
• Exclude known transient folders if they are not part of the migration scope
• Run the old and new exports separately and compare later

17. Final recommendation

For CIFS/SMB migration validation, do not rely on only one view. The safest approach is:
• compare **inventory** for missing data
• compare **NTFS ACLs** for file/folder permissions
• compare **share ACLs** for the SMB share-level access

That combination gives you a practical sign-off report for old vs new CIFS shares.